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TEACHING CASE

Automation in recruitment: a new frontier

Pooja Gupta[1,2] · **Semila F. Fernandes**[1,2] · **Manish Jain**[1,2]

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Abstract The case is about the urge for improving the processes and utilizing technologies available in market for Human Resource Management Services especially in recruitment. The dilemma in the case is whether the company should opt for automation or not. And if the company is adopting automation to support workforce planning and recruitment functions, is the company ready for the same. The HR industry at large is aware of the challenges faced by the recruitment teams in terms of the various hiring functions involving sales engine, sourcing, interviewing, providing offers, post offer follow-ups and finally joining and induction. To address this challenge, the case touches on instituting a culture change by embracing technology into divisions which are not yet open to adopting innovative methods. The company is looking at incorporating Artificial Intelligence in its various processes. It is also looking at Robot Process Automation to perform standard HR functions in the recruitment process. The company must decide whether to implement automation and if it does, how it should be executed. The case also debates on whether technology would increase the effectiveness of talent acquisition function and bring a change in the administrative HR image or would this automation-led hiring take away the control from the company human resources?

Keywords Artificial Intelligence · Recruitment · Robotic process automation · Technology

Manav looked at his watch. The meeting was due to start in five minutes. Manav was the director of Recruitment Services in a leading Human Resource service company in India. The meeting was very crucial for Manav as it was an outcome of an idea promulgated by him about incorporation of technology in HR practices especially recruitment.

Manav had been toying with the idea of using automation in recruitment processes for some time. He had not been sure on how to go about it. One day, when it was raining heavily, he was forced to share a cab with Manish Jain. Manish was a project manager in a mid-size IT company. The two got talking about technology and Manav's idea came to forefront.

Manish was very enthused about the idea of incorporating technology in HR services and wanted to further work on the idea. He was the head of Automation division in his company. His division was currently working on automation of services in the IT sector. When he heard Manav's ideas, he realized that the same principles of automation and technology use would be applicable in HR services industry as they were doing in other industries right now.

The very next day, Manav, spoke to the CEO of his company, Pranav Singhvi, regarding his idea and his discussions with Manish. Pranav was intrigued by the idea of automation in recruitment and agreed to invite Manish and his team for discussions on learning more about how to use technology and automation in recruitment processes.

Teaching Notes can be obtained from the corresponding author.

✉ Pooja Gupta
poojagupta@sibm.edu.in

Semila F. Fernandes
semila.fernandes@sibm.edu.in

Manish Jain
jainmani2602@yahoo.com

[1] Symbiosis Institute of Business Management, Bengaluru, India

[2] Symbiosis International University, Pune, India

Manav asked Manish to submit a formal proposal regarding the automation of the processes in recruitment. Today's meeting was the first presentation of the proposal. Manav was eagerly looking forward to hear the plan of Manish's team.

The company—HR solutions[1]

HR solutions was established in the year 2000. Since then, it has become India's leading human resource service companies in the organized sector. It was a private limited company with offices in 12 major cities of the country along with more than 1500 clients all over India. Today it is seen as a one-stop provider of human resources services to various industries and diverse functional roles, offering staffing, payroll processing, recruitment, compliance, and training services.

The company provides services in the following areas:

- The company assists its clients in their Regular or Seasonal staffing requirements by providing contractual or temporary staffing services. The company has emerged as one of the India's largest temp staffing companies, with 1000+ temporary staffing clients and an outsourced employee base of over 100,000+ associates.
- IT staffing—the company has built an expertise in the area of IT resource requirements. The company has over 15 years of expertise in the IT staffing industry. With an integrated IT Staffing capability, the experience of the company adds value to our clients' talent acquisition goals.
- Payroll processing—the company has a well-established and robust payroll processing system in place that is a combination of the best of software tools, expert know how, IT infrastructure, and a strong payroll team.
- Recruitment services—the company offers permanent recruitment solutions to over 800 active clients across the country. They bring to the table an experienced team of recruitment professionals, with domain specialization to cater to specific industries. They provide clients with the right solution to finding key people faster, while freeing up management bandwidth to concentrate on core management activities.
- Recruitment process outsourcing—the company is amongst the top recruitment outsourcing companies in India, enabling their partner organizations with recruitment process outsourcing solutions designed to understand the very nature of the client's problem. The RPO model reduces the recruitment complexity for the clients; bringing predictability to hiring, optimizing recruitment costs, improving candidate experience, and enhancing talent quality for the clients ensuring the right people, with the right skillsets are available at the right time.

The challenges

Manav, after the go ahead from Pranav, started collating information regarding the recruitment processes and the major pain points. One of the things which came up repeatedly was lethargy in manual processes and dependencies. The dependencies maybe due to other departments, other agencies, and the uncertainty regarding candidate decision to join or not. It was impacting overall performance of company.

During this time, Manav came across several industry reports such as "*Employment Outlook Report*[2]" from TeamLease™ published for 2017, "*Outlook on Artificial Intelligence in the Enterprise 2016*[3]" and the *PwC HR technology Survey, 2017.*[4]

The "Employment Outlook report" from TeamLease, spoke about, "6% points dip, for the forthcoming half year, April—September, 2017" hinting us that the market is challenging. At the same time, demands from tier 2 cities are high." It also spoke about the tilt towards BFSI, Healthcare & Pharma and KPO as they have high hiring capacities and also they are top sectors in employment outlook growths. This report led towards demand of resolving challenges and improving the ROI by streamlining hiring processes and interaction with clients.

"Outlook on Artificial Intelligence in the Enterprise 2016" highlighted 2 major outcomes:

1. Between 2014 and 2015 alone, for example, the number of organizations either deploying or implementing DATA-DRIVEN PROJECTS INCREASED BY 125%, with the average enterprise spending $13.8 million on the effort.
2. Market intelligence firm IDG also projects that the market for BIG DATA TECHNOLOGY AND SERVICES WILL REACH $48.6 BILLION by 2019.

Artificial intelligence had been available in the market for more than two decades, but its application had been

[1] Name withheld on request of the company.

[2] www.teamlease.com/sites/default/files/resources/employment_outlook_report.pdf.

[3] www.datascienceassn.org/…/Outlookpercent20onpercent20Artificialpercent20Intelligencepercent20inpercent20t….

[4] https://www.pwc.com/us/en/hr…/technology/global-hr-technology-survey.html.

mostly on mission critical projects and medical equipment industries. According to the report, 61% of the "innovation strategy" respondents were applying AI to their data to identify previously unidentified opportunities like process improvements or new revenue streams while only 22% of respondents were trying to take advantage of this opportunity without a proper strategy in place.

All the Artificial Intelligence ecosystems mentioned below are not only penetrating in all businesses but also brining in many new opportunities.

Evidence based	Machine learning	Prescriptive analytics
Deep learning	NLP/text mining	Predictive analytics
Recommendation engines	Natural language generator	

Manav thought as to why Artificial intelligence be not applied to Human Resource Management.

He also came across PwC's HR Technology Survey which was an interesting complement on the above two reports. They highlighted cloud adoption and analytic applications prioritization by the chief human resource officer's agenda. Two years ago, 68% of organizations had at least one HR process in the cloud. In 2017, that number has climbed to 73%. As a further barometer, nearly 40% have their core applications in the cloud. Another interesting fact came out in this report is 20% of companies saw a decrease in HR personnel costs with their move to the cloud given the increased automation and innovation.

Manav highlighted the following stages in recruitment to help Manish and his team in understanding the recruitment process.

- *Requirements planning*—as the requirement arises for staffing, the requirement gathering phase kicks in. Manav explained that recruitment manager needs to engage the Hiring manager multiple times to collect and understand the requirements. These visits at times can be very time consuming due to different locations of the recruitment and the hiring manager. However, in all requirements, 90% of the time the questions asked and data points discussed are same. The recruitment manager's face-to-face meetings with the hiring managers were mostly about creating long-term relationship and seeking further opportunities.
- *Sourcing*—in spite of having many social media channels available in today's world, the information is not enough for sourcing and selecting the resumes. Study says though the candidate has passed his interviews, he was rejected on the basis of his social media posts. Referrals play a major role in sourcing.
- *Interviewing*—every client has demands of different levels of interview. The information technology clients will vary interview from qualitative questionnaires, telephonic, and face to face with case demonstration. This also creates high pressure on candidates as well as recruitment managers. No show for interview is the most common problem faced by recruitment managers. Hiring Manager's non-availability even though the interview was planned well in advance is another issue. This piles up multiple channels of communication and coordination.
- *Offer letter*—Many companies are now adopting the digital media for processes. The recruitment manager needs to keep the candidate warm enough till the time he joins. Candidates need to be engaged by providing new company's details and work environment.

The survey

In order to decide on the automation and use of technology in recruitment process, Manav floated a survey for the recruiting managers. The questionnaire was designed based on requirements for volume hiring. As he wanted the survey to substantiate his usage of technology theory, he mentioned the purpose of each section of the survey. The questions and their purpose are given as follows:

Hiring stage	Questions	The purpose of the question used
Requirements planning	Interview with hiring manager provides more details for JD Hiring manager's inputs via Video conferencing contributes in better planning understanding A template for requirement will improve the efficiency of Sourcing candidate resumes Automated process enables better requirement gathering	The questions were planned to get the recruiters view on usage of tools and technologies like interviewing, video conferencing etc. This was done to help in deciding the tools or technologies to be used in requirement planning stage of hiring and enable recruiters to save their time and utilize the time in better and effective requirement planning
Sourcing	Social Media provides enough sourcing of candidates resumes	The question was planned to get the recruiters assessments and provide their options on usage of latest technologies like Social media, web-based search engines etc



Hiring stage	Questions	The purpose of the question used
	Social media like Facebook, Twitter, Glassdoor and WhatsApp provide additional details of candidates as inputs for sourcing decisions	
	While sourcing, additional information of candidate needs to be collected so that we can track the candidates post offer	
	Due to Social media, there are possibilities of losing out other best/talented candidates for sourcing	
	Technology enablement like optimized web-based search engine assists in quality of sourcing	
Interviewing	Qualitative questionnaires will yield better result in interviewing process	The questions were planned to get the recruiters to evaluate options on usage of latest technologies like Interactive Voice Response calls, video conferencing for interviewing etc
	ATS (Application Tracking System) with qualitative feedback for recruiter improves and enables Hiring/ Sourcing process	
	IVR calls to candidate and hiring manager like 1 day before and 5 min before helps in curtailing no show	
	Automated online interview assessment increases the efficiency of decision making for the hiring managers	
	Video interviewing saves time, effort cost and travel	
Offer Letter	Online data collection is the solution for offer and background verification	The questions were planned to persuade the recruiters to consider options on usage of latest technologies like Digitization, Phone calls, video conferencing for offer letter explanation etc
	Digital offer letter provides a time savings for candidate as well as recruiter, such as travel and convenience	
	Offer letter walk through over phone or video is effective mechanism	
	Offer formalities using online medium or phone calls are more effective methods experienced by Candidates	
	End-to-end customized candidate email templates and automated candidate tracking/engagement workflows benefits hiring process	
Post offer Follow up (POFU)	Candidate engagement through mobile platform creates interest for the candidate to join the organization without referring to other opportunities	The questions were planned to get the recruiters to gauge options on usage of latest technologies like mobile platform, Instructor Led Training etc
	The Hiring manager dashboard accurately predicts the intent of the candidate to join the organization through virtual mobile platforms	
	The recruiter understands that the candidate drop ratio for the offer or not join the company he/she is offered on the date of joining	
	The candidate feels more engaged if the company offers insights in there EVP's, career path, Engagements, Work ethics and other insiders through mobile or web-based platform	
	Online or Instructor led training (ILT)-based induction formalities are very effective	

Hiring stage	Questions	The purpose of the question used
Technology	With Technology, recruiting is potentially Better, Faster and Cheaper Technology enables the understanding on the longevity of the employees Technology usage in hiring improves "Hiring cost" Technology usage in hiring improves "Quality of hire" metrics Technology usage in hiring improves "Competency/skills gap analysis" metrics Technology usage in hiring improves "Professional development" metrics Technology usages in hiring loses out on some significant candidates Technology usage in hiring improves the transparency to all stakeholders	The questions were planned to get the recruiters to weigh up options on usage of latest technologies. The aim was to identify tools or technologies that will engage & enable RPO recruiting team to contribute in improving efficiency in process. This would in turn improve the cost of hiring process and contribute in better revenue per hire

The survey results are shared in Appendix 1. Manav also shared the survey results with Manish.

The presentation

Manish started his presentation with a quote from Lacity and Wilcocks,[5] "In the guise of organizational efficiency, managers have been trying to get humans to act as robots by structuring, routinizing and measuring work. The automation software being developed in today's time, is enabling the reversal of this process. Companies today are using software and automation to amplify and augment distinctive human strengths, enabling large economic gains and more satisfying work."

Manish spoke about the pervasiveness of Artificial Intelligence in today's corporate world. He defined artificial intelligence very broadly in terms of computers being able to perform tasks that once required human intelligence. He used literature[6] to highlight the fact that the utility of artificial intelligence should be seen from the point of view of business capabilities rather than the technologies. He highlighted three different and important needs which could be supported by artificial intelligence. These needs were defined as follows:

- Robot process automation (RPA)—This includes the automation of digital and physical tasks using robots. It is the least expensive and easiest to implement of the cognitive technologies. It is found to be particularly well suited to multiple back end systems.
- Cognitive insights—this uses algorithms to detect patterns in vast volumes of data and interpret their meaning. The insights provided through machine learning are found to be more enriching as they are more data intensive and detailed in nature.
- Cognitive engagement—These include tools such as natural language processing chatbots, intelligent agents and machine learning to engage customers, current employees and prospective employees.

Manish then went on to make a presentation on the integration of technology in the hiring process and automation of several routine processes. The main points highlighted by him were as follows:

- It is the right time to automate several processes in recruitment and hiring
- Usage of video conferencing as much as possible, starting with meetings between hiring manager and the recruiting manager, initial interviews with outstation candidates, and onboarding processes.
- Usage of social media to verify and screen the candidates. The social media screening can be at various levels and may need to go in depth at times when required.
- Auto-reminders to be used at various levels of recruitment
- Special apps to engage the candidate once he/she is hired
- Recruitment dashboard for hiring and recruiting managers for checking the hiring status
- Online onboarding and training

He also gave a brief about the kind of automation tools currently being used by other companies across the world in recruitment functions.

- Innovative learning tools like app-based games, simulation exercises: which provide insights on performance

[5] Lacity, M. C., & Willcocks, L. P. (2016). A new approach to automating services. *MIT Sloan Management Review*, *58*(1), 41.

[6] Davenport, T. H., & Ronanki, R. (2018). Artificial Intelligence for the Real World. *HARVARD BUSINESS REVIEW*, *96*(1), 108–116.



  - They would support in identifying global talent in a collaborative learning environment.
  - app-based video games provide the ability to judge the candidate's psyche, intellect, ability to work as a leader and an innovator etc. where-in the applicant's behavior is studied from 20 min on the game being played.
- Predicative analytics: They support the HR function where-in the educational qualifications of the applicant's along with the performances of the tests are fed into the database which are correlated for huge number of hires.
- Gamification processes: that engages potential hires and socializes them to the organizational culture to increase the candidates fit.
  - This data analytics technique increases engagement levels of the potential future employees and identifies the development needs of all its members.
- Organizations adopted iTAP Ideapreneurship talent acquisition platform: it provides a platform to look into the culture of the candidates, track and report performance based on this new performance metric.
  - iTAP also supported the functionality of publishing regular reports on the parameters which are key for the business along with providing week-on-week engagement with the candidates and providing insights on the probability of the candidates joining the company.
- Use of mobile devices and cloud-friendly systems: It will empower working from dispersed locations
- Engagement models: which include outsourcing, crowd sourcing, and freelance works would be looked for by companies in the future

Manish believed technology usage would help in

(a) Recruiting faster
(b) Optimize the hiring cost,
(c) Quality of hire
(d) Develops competency and skills for recruiting managers
(e) Improves professional development for recruiting managers
(f) Brings more transparencies

The Dilemma

The CEO of the company thanked Manish and his team for the presentation. There were still some doubts persisting in his mind. He invited Manish and Manav for a private meeting and shared his apprehensions:

- Will the clients be as ready to embrace the latest skill sets in technology?
- Is the company ready for the impact and implications of automation?
- Would the implementation of automation bring new opportunities for the company?
- Would Human Resource Management lose the Human touch?
- Is video conferencing same as face-to-face meeting with clients? Would it have any impact on the results?
- Automation of processes may expedite the process but can it be applied unilaterally?
- Would the employees of HR solutions be threatened by the technology?

Exhibit 1: List of Readings

- Lacity, M. C., & Willcocks, L. P. (2016). A new approach to automating services. *MIT Sloan Management Review*, *58*(1), 41.
- Davenport, T. H., & Ronanki, R. (2018). Artificial Intelligence for the Real World. *HARVARD BUSINESS REVIEW*, *96*(1), 108–116.
- Wilson, H. J., Daugherty, P., & Bianzino, N. (2017). The jobs that artificial intelligence will create. *MIT Sloan Management Review*, *58*(4), 14.
- Maurer, S. D., & Liu, Y. (2007). Developing effective e-recruiting websites: Insights for managers from marketers. *Business horizons*, *50*(4), 305–314.

Appendix: Data analysis and survey results

The main survey was collected from a group of 52 recruiters.

Requirements planning responses summary

Eighty five percentage of recruiters would like to interview with hiring manager for better understanding of the Job description and gauge the hiring manager's interest. 62% of recruiters feel use of video conferencing is better means for interviewing Hiring manager and it will help them in efficient requirement planning. Again, 59% of recruiters feel a defined template will bring added advantage and contribute in sourcing. This will also improve the efficiency of sourcing candidate resumes and enhance service. 49% of recruiter concur that automated process will enable better requirement gathering.

Sourcing responses summary

Though 25% of recruiters believe that Social media is good place for sourcing resumes. However, 50% of recruiters neither agree nor disagree. 48% of recruiters agree that other Social media like Facebook, Twitter, Glassdoor, WhatsApp etc. provide additional details of candidates. These media are facilitating in sourcing decisions. 94% of recruiters agree that additional information of candidates, collected from Social Media, helps in tracking the candidates post offer. Though there are passive candidates who may not be on Social Media, 27% of recruiters feel the possibilities of losing the talented candidates. 71% of recruiters agree on technology enablement like optimized web-based search engine will improve the quality of sourcing.

Interviewing responses summary

Seventy five percentage of recruiters agree that Qualitative questionnaires will yield better result in interviewing process. 81% recruiters subscribe the thought of ATS (Application Tracking System) with qualitative feedback will improve and enable Hiring/Sourcing process for them. 63% of recruiters also agree that IVR calls to candidate and hiring manager helps in curtailing No Show. 58% of recruiters agree and subscribe to the automated online interview assessment will increase the efficiency of decision making for the hiring managers. 89% of recruiters agree on video conference-based interviewing will save time, effort cost, and travel.

Offer letter responses summary

Fifty percentage of recruiters agree that online data collection is the solution for offer and background verification. 83% of recruiters concur with the idea of Digital offer letter will provide a time savings for candidate as well as recruiter, such as travel and convenience. 2% disagree with digitization. 50% of recruiters agree that Offer letter walk through over phone or video will be effective mechanism. However, 19% of recruiters disagree with the thought of Offer letter walk through over phone or video will be effective mechanism. 65% of recruiters agree with Offer formalities using online medium or phone calls are more effective methods experienced by Candidates. Though, 11% of recruiters disagree with this concept, 81% of recruiters agree with end-to-end customized candidate email templates and automated candidate tracking/engagement workflows will benefit hiring process.

Post offer follow-up (POFU) responses summary

Fifty percentage of recruiters agree with candidate engagement through mobile platform will creates interest for the candidate to join the organization without referring to other opportunities. 31% of recruiters have mixed feeling on this thought. 42% of recruiters agree on Hiring manager dashboard that accurately predicts the intent of the candidate to join the organization through virtual mobile platforms. However, 44% of recruiters have mixed response on Hiring manager dashboard that accurately predicts the intent of the candidate to join the organization through virtual mobile platforms. 48% of recruiters understand that the candidate drop ratio for the offer or not joining the company he/she is offered on the date of joining. 85% of recruiters agree that the candidate feels more engaged if the company offers insights in there EVP's, career path, Engagements, Work ethics, and other insiders through mobile or web-based platform. 4% of recruiters disagree that the candidate feels more engaged if the company offers insights in there EVP's, career path, Engagements, Work ethics, and other insiders through mobile or web-based platform. 48% of recruiters agree that Online or Instructor led training (ILT)-based induction formalities are very effective. However, 27% of recruiters neither agree nor disagree that Online or Instructor led training (ILT)–based induction formalities are very effective. Interestingly, 25% of recruiters disagree that Online or Instructor led training (ILT)-based induction formalities are very effective.

Technology responses summary

Ninety six percentage of recruiters agree that with Technology, recruiting is potentially Better, Faster, and Cheaper. 50% of recruiters agree that Technology enables the understanding on the longevity of the employees. 39% of recruiters have mixed feeling about Technology enables the understanding on the longevity of the employees. 77% of recruiters agree that Technology usage in hiring improves "Hiring cost." 79% of recruiters agree that Technology usage in hiring improves "Quality of hire" metrics. 75% of recruiters agree that Technology usage in hiring improves "Competency/skills gap analysis" metrics. 73% of recruiters agree that Technology usage in hiring improves "Professional development" metrics. 35% of recruiters agree that Technology usages in hiring will miss significant candidates. However, 44% of recruiters neither agree nor disagree that Technology usages in hiring lose out on significant candidates. 65% of recruiters agree that Technology usage in hiring improves the transparency to all stakeholders.



Pooja Gupta is a BA (Hons.) Economics Graduate from Sri Ram College of Commerce, University of Delhi. She has done her MBA Finance from International Management Institute New Delhi. She is currently doing PhD in the area of Finance. She has more than 11 years of teaching experience and industry experience of 1 year. She teaches Corporate Finance, Security Analysis, Valuation and International Business. Her research interest is in the area of Contrarian Investing and Behavioural Finance. She has attended and presented papers in various International and National Conferences.

Semila F. Fernandes holds an MBA in Marketing Management from Pune University, M.Phil in Management from Periyar University, and B.Sc from Goa University. She has also completed "Post graduate Diploma in Intellectual Property Rights" from NLSIU, Bangalore. Presently she is pursuing her Ph.D from SIU Pune. Her teaching experience consists of 13 years in the area of Marketing. Prof. Semila has also got corporate exposure at Bajaj Allianz General Insurance Company as Executive—Direct Marketing and at ICFAI as Regional Summer Internship Co-ordinator and has been actively involved in conducting workshops and orientation for prospective MBA students. Her research expertise has been in the area of Consumer Behavior.

Manish Jain is an Engineer and MBA with overall 15 years of proven experience in delivering large-scale complex engagements. Managed budgets of $8 M with GM of 18%. He is a keen strategist, planner & implementer with expertise in establishing & managing entire delivery with key focus on bottom line profitability by ensuring optimal utilization of resources, E2E solution integrity.

